

Karnataka Chemistry - 2015

General Instructions:

- (1) The question paper has 4 parts. All parts are compulsory.
- (2) Part-A carries 10 marks. Each question carries one marks.
Part-B carries 10 marks. Each question carries two marks.
Part-C carries 15 marks. Each question carries three marks.
Part-D carries 35 marks. Each question carries five marks.
- (3) Write balanced chemical equations and draw neat diagrams wherever necessary.
- (4) Use log tables and simple calculator if necessary. (Use of scientific calculator is not allowed.)

PART-A

Answer all the question each question carries one mark. (Answer each question in one word or one sentences:

1. At a given temperature and pressure Nitrogen gas is more soluble in water than helium gas which one of them has higher value of K_H ?

Sol. Helium gas has higher value of K_H .

2. On mixing equal volumes of acetone and ethanol, what type of derivation from Raoult's law is expected?

Sol. On mixing equal volumes of acetone and ethanol, Positive derivation from Raoult's law is expected.

3. What happens to molar conductivity when one mole of KCl dissolved in one liter is diluted to five liters?

Sol. Molar conductivity increases when one mole of KCl is dissolved in one liter is diluted to five liter.

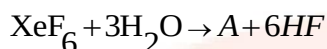
4. What happens to the half-life period for a first order reaction, if the initial concentration of the reactant is increased?

Sol. Half-life period of a first order reaction remains unchanged, if the initial concentration of the reactant is increased.

5. Write the process usually employed for the purification of Nickel.

Sol. Mond's process based on vapors phase refining is usually employed for the purification of nickel.

6. Identify the product 'A' in the following reaction



Sol. $\text{XeF}_6 + 3\text{H}_2\text{O} \longrightarrow +6\text{HF}$

$\therefore \text{XeO}_3$ is A

7. How many moles of AgCl will be precipitated when an excess of AgNO_3 solution is added to one molar solution of $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2$?

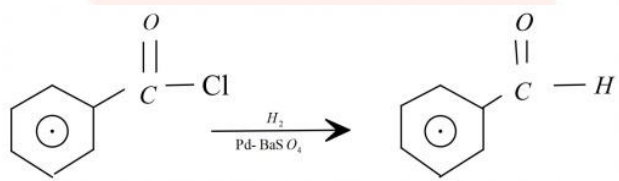
Sol. Two moles of AgCl will be precipitated when an excess of AgNO_3 solution is added to one molar solution of $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2$.

8. Name the organic product formed when chlorobenzene is treated with sodium in dry ether.

Sol. When chlorobenzene is treated with sodium in dry ether, biphenyl/diphenyl is found

i.e., Fig

9.



Name the above reaction.

Sol. This is Rosenmund reaction for the preparation of aldehydes.

10. Deficiency of which vitamins causes the disease pernicious anemia?

Sol. Deficiency of vitamin B_{12} causes pernicious anemia.

PART-B

Answer any five of the following. Each question carries two marks.

11. What is meant by the term condition number in solids? What is the coordinate number in a face centered cubic close packing structure?

Sol. Co-ordinate number in solids is the number of nearest neighbouring particles surrounding the given particle. The co-ordinate number in a face centered cubic close packing structure is 12.

12. State Faraday's law of electrolysis. For the electrode reaction $Zn^{2+} + 2e \rightarrow Zn_{(s)}$, What quantity of electricity in coulomb's required to deposit one mole of zinc.

Sol. Faraday's first law of electrolysis:-

The amount of chemical reaction (amount of substance liberated or deposited) at each electrode during electrolysis by a current is directly propotional to the quantity of electricity passed through the electrolyte (melt or solution).

For the reaction $Zn^{2+} + 2e \rightarrow Zn$. Quantity of electricity in coulomb's required to deposit one mole of zinc is 2 Faraday or $2 \times 96500C = 193000C$.

13. A reaction is first order with respect to the reactant A and second order with respect to the reactant B in a reaction $A + B \rightarrow \text{product}$.

(i) Write the differential rate equation.

(ii) How is the rate of the reaction affected on increasing the concentration of B by two times?

Sol. (i) Differential rate equation is

$$r = k[A]^1[B]^2$$

Where r is rate of reaction k is rate constant.

(ii) The rate of the reaction increases to 4 times the initial rate on increasing the concentration of B by two times.

14. Give a two differences between lanthanoids and actinoids.

Sol. Lanthanoids:

- Except promethium, all the other lanthanoids are non-radioactive. In addition to +3 oxidation state, -they show +2 and +4 state.

- Their magnetic properties can be studied easily.
- They show less tendency to form complex compounds.
- They do not form oxocations.

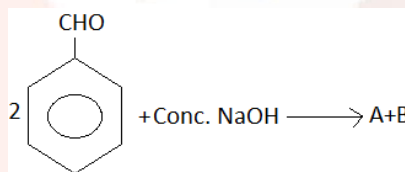
Actinoids :

- All the actinoid are radioactive.
- In addition to +3 oxidation state, they show +4,+5,+6,+7 oxidation state.
- Their magnetic properties are difficult to study.
- They have more tendency to form complex compounds.
- They form oxidations like UO_2^{2+} , UO^+ , PuO_2^{2+} to.

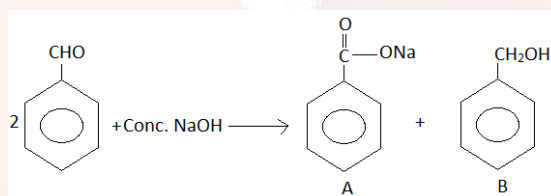
15. Name the product formed when phenol is treated with acidified solution of $\text{Na}_2\text{Cr}_2\text{O}_7$.
Give equation.

Sol.

16. Identify A and B in the following reaction:



Sol.



A is sodium benzoate.

B is benzylalcohol.

17. What is the role of these as food additives?

(i) Sodium benzoate

(ii) Aspartame.

18. Explain saponification of oils/fats with equation.

PART-C

Answer any five of the following. Each question carries three marks.

19. Describe the three steps involved in the leaching of Bauxite to get pure Alumina (equations not expected).

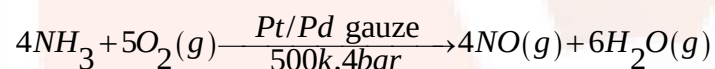
Sol. Leaching of Bauxite:

Bauxite, the chief ore of aluminium is leached with concentrated NaOH solution at 473K-523K temperature and a pressure of 35-36 bar. Bauxite is leached out as sodium aluminate and silica as sodium silicate leaving behind the impurities.

20. Write the equations involved in the preparation of Nitric acid by Ostwald's process by maintaining the reaction conditions.

Sol. Preparation of nitric acid by Ostwald's process. It involves three steps.

-Ammonia is oxidized by atmospheric oxygen in the presence of a catalyst to form nitric acid.

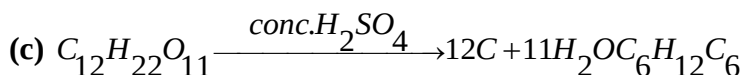
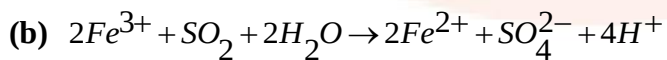
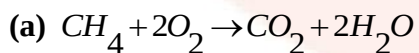


-Nitric oxide is further oxidized to form nitrogen dioxide. i.e. $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$.

-Nitrogen dioxide is dissolved in water to form nitric acid. i.e.

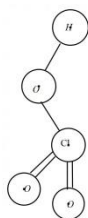
$3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO(g)$ Thus, nitric acid is obtained and NO is regenerated which can be reversed in second step nitric acid can be concentrated by distillation upto 68% by mass.

21. Complete the following equations:



22. (a) Which is the strongest acid among the hydrogen halides? Give one reason [X=F, Cl, Br, I].

(b) Write the structure of chloric acid (HClO_3).



Sol. (a) The strongest acid among hydrogen halides is hydrogen iodide (H-I), this is because bond-dissociation enthalpy of H-I is least among other hydrogen halides. As H-I bond is weak due to long bond length.

(b) Structure of HClO_3 .

23. Give reason (one each) for the following:

(a) Transition metals are good catalytic agent.

(b) Second ionization enthalpy of copper are very high.

(c) The spin only magnetic moment of Sc^{3+} is zero ($Z=21$).

Sol. (a) Transition metals are good catalytic agent because:-

- They provide large surface area for adsorption.
- they exhibit variable oxidation states.
- They have a tendency to form intermediate complex compounds.

(b) second ionization enthalpy of copper is very high due to the stable $3d^{10}$ configuration of Cu^+ (i.e., $[\text{Ar}]3d^{10}$).

(c) The spin only magnetic moment of Sc^{3+} is zero due to the absence of unpaired electrons in both 3d and 4s orbitals.

24. Write the equations involved in the preparation of potassium Dichromate from chromite ore.

25. With the help of valence bond theory account for hybridization, geometry and magnetic property of $[\text{Ni}(\text{CN})_4]^{2-}$ complex ion [Z for Ni=28].

26. (a) For the given complex $[Co(NH_3)_5Br]SO_4$, write the IUPAC name and its ionization isomer.

(b) Which set of d-orbitals of metal ion/atom experience more repulsion in octahedral field created by the ligand?

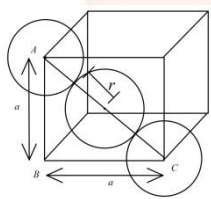
PART-D

Answer any three of the following. Each question carries five marks.

27. (a) Calculate the packing efficiency in a unit cell of cubic close packing (CCP) structure.

(b) Name the crystal defect which lowers the density in an ionic crystal.

Sol. (a) Consider a unit cell (CCP) with edge length 'a'.



Let the radius of each particle be 'r' in CCP, particles touch each along the face diagonal. Let $AC = \text{face diagonal} = b$

From $\triangle ABC$ (right angled)

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow b^2 = a^2 + a^2$$

$$\Rightarrow 2a^2$$

$$b = \sqrt{2}a \dots\dots (1)$$

But from the diagram, $b = 4r \dots\dots (2)$

Equating (1)&(2), We get

$$\sqrt{2}a = 4r$$

$$\Rightarrow a = \frac{4r}{\sqrt{2}}$$

$$\Rightarrow a = 2\sqrt{2}r$$

Now volume of unit cell $= a^3$

$$= (2\sqrt{2}r)^3$$

$$= 16\sqrt{2}r^3$$

There are four particles per unit cell of CCP volume of each particle $= \frac{4}{3}\pi r^3$

So, volume of 4 particles $= 4 \times \frac{4}{3}\pi r^3$

$\frac{16}{3}\pi r^3$ cubic units.

Packing efficiency of cubic close packing (CCP)

Volume occupied by all 4 particles $\times 100$

Volume of unit cell, $\frac{\frac{16}{3}\pi r^3}{16\sqrt{2}r^3} \times 100$

$$= \frac{\pi}{3\sqrt{2}} \times 100$$

$$= 0.74 \times 100$$

$$= 74\%$$

Therefore Packing efficiency in CCP = 74%

(b) Schottky defect lowers the density in an ionic crystal.

28.(a) A solution containing 18g of non-volatile non-electrolyte solute is dissolved in 200g of water freezes at 272.07K. Calculate the molecular mass of solute. Given: $K_f = 1.86K$ Kg/mol. Freezing point of water = 273K

(b) Define isotonic solution. What happens when the blood cell is dipped in a solution containing more than saline concentration?

Ans(a). Mass of solute $(w_2) = 18g$

Mass of solvent $(w_1) = 200g$

Kg of solvent $= 1.86K$ Kg/mol

Molar mass of solvent $(M_1) = 18g$

Depression in freezing point.

$$\Delta T_f = 273K - 272.07K$$

$$\Delta T_f = 0.93K$$

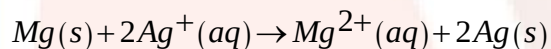
Molecular mass of solute i.e.

$$\begin{aligned} M_2 &= \frac{1000K_f W_2}{\Delta T_f W_1} \\ &= \frac{1000 \times 1.86 K K g / mol \times 18g}{0.93K \times 200g} \\ &= \frac{33480}{186} g / mol \\ &= 180g / mol \end{aligned}$$

Therefore molecular mass of solute = 180g/mol

(b) A solution which has same osmotic pressure as that of given solution is said to be isotonic solution. When blood cell is dipped in a solution containing more than normal saline concentration, water flows out of the cell and the cell shrinks.

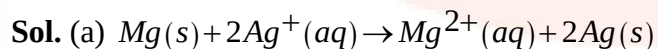
29. (a) Calculate the EMF of the cell for the reaction.



Given: $E^0_{Mg^{2+} / Mg} = -2.37V$

$$\begin{aligned} E_{cell} &= E^0_{cell} - \frac{2.303RT}{nF} \log \left(\frac{[Mg^{2+}]}{[Ag^+]^2} \right) \\ &= 3.17 - \frac{2.303 \times 8.314 \times 298}{n \times 96500} \log_{10} \left(\frac{0.001}{0.0001} \right) \\ &= 3.17 - \frac{0.0591}{2} \log_{10} 10 \\ &= 3.17 - 0.02955 \times 1 \\ &= 3.14045V \end{aligned}$$

(b) What are fuel cells?



Given: $E^0_{Mg^{2+} / Mg} = -2.37V$

$$E^0_{Ag^+ / Ag} = 0.80V$$

$$[Mg^{2+}] = 0.001M; [Ag^+] = 0.0001M$$

$$\log 10^5 = 5$$

$$\begin{aligned} E^0_{cell} &= E^0_{cathode} - E^0_{anode} (= E^0_{RHE} - E^0_{LHE}). \\ &= 0.80 - (-2.37) \\ &= 3.17V \end{aligned}$$

From nearest equation,

$$\begin{aligned} E_{cell} &= E^0_{cell} - \frac{2.303RT}{nF} \log \left(\frac{[Mg^{2+}]}{[Ag^+]^2} \right) \\ &= 3.17 - \frac{2.303 \times 8.314 \times 298}{n \times 96500} \log_{10} \left(\frac{0.001}{0.0001} \right) \\ &= 3.17 - \frac{0.0591}{2} \log 10 \\ &= 3.17 - 0.02955 \times 1 \\ &= 3.14045V \end{aligned}$$

(b) Fuel cells are the galvanic cells (electrochemical cells) used to convert the energy of combustion of fuels like methane, hydrogen etc, into electrical energy.

30. (a) Derive integrated rate equation for the first order reaction.

(b) According to collision theory, what are the two factors that lead to effective collisions?

31. (a) Write any two differences between physisorption and chemisorption.

(b) Name the phenomenon/effect for the following:

(i) Colloidal particles are in Zig-Zag motion.

(ii) When an electrical potential is applied across two platinum electrodes dipping in a colloidal solution, particles move towards one or the other electrodes.

(iii) Scattering of light by colloidal sol.

Answer any four of the following. Each question carries five marks.

32. (a) Write equations for the steps in SN^1 mechanism of the conversion of tert.butyl bromide into tert.butyl alcohol.

(b) Identify the products A, B and C in the following equation.



Sol. (a) SN^1 mechanism:-

-Usually, SN^1 reactions are carried out in polar protic solvents such as water, alcohol etc,

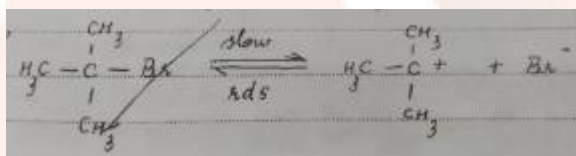
-Here the rate of reaction depends on the concentration of only one reactant.

-It involves 2 steps:-

Eg: Conversion of tert-butyl bromide into tert-butyl alcohol.

Step1:

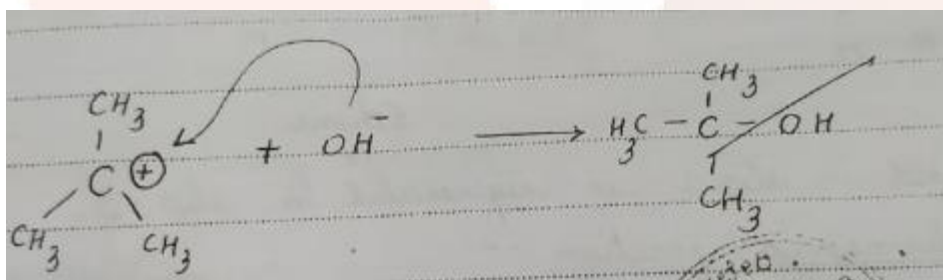
Tertiary butyl bromide undergoes slow ionization to form tertiary carbocation.



i.e.

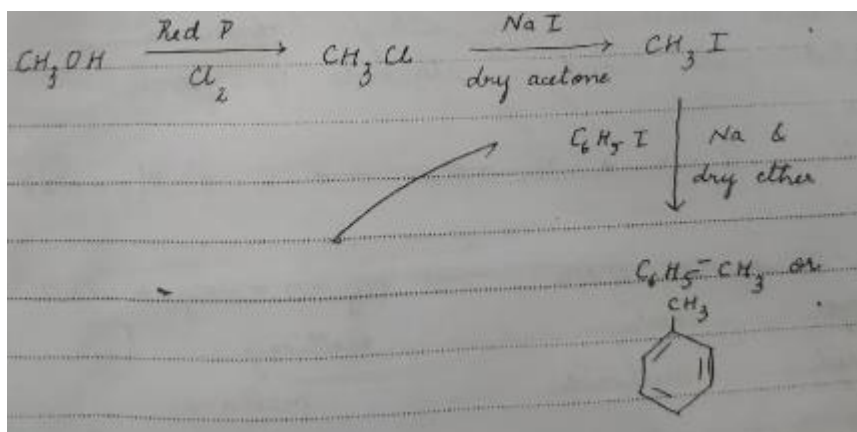
Step-2:

Tertiary carbocation thus formed is planar since it is sp^2 hybridized therefore, the nucleophile OH^- attacks the carbocation from any direction to form tert-butyl alcohol.



i.e.

$$r = k [\text{tert-butyl bromide}]$$



(b)

33. (a) Write the mechanism of acid catalyzed dehydration of ethanol to ethane.

(b) Explain Williamson's reaction. Write the general equation.

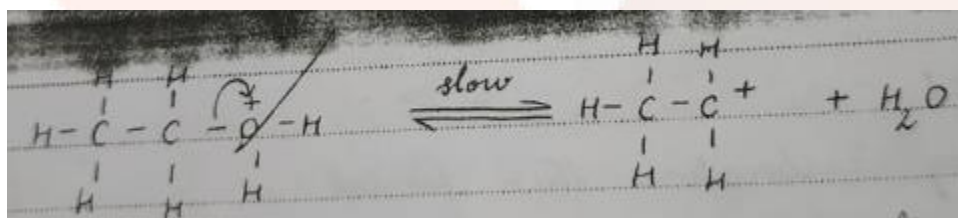
Sol. (a) Acid catalysed dehydration of ethanol is treated with conc. H_2SO_4 at 443K, dehydration takes place to form ethane.

This takes place in three steps:

(i) Formation of protonated alcohol.

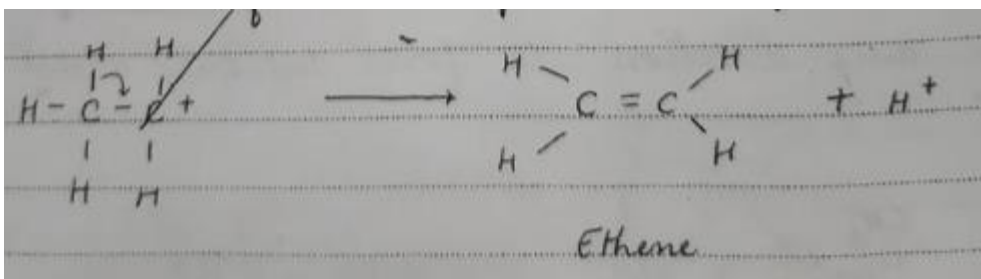


(ii) Formation of carbocation



It is the slowest step and hence rate determining step.

(iii) Formation of ethene by removal of proton:



Acid used in step-1 is regenerated in step-3.

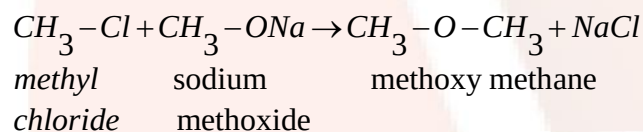
(b) Williamson's reaction:

This is an important laboratory method for the preparation of both symmetrical and unsymmetrical ethers.

-In this method, an alkyl halide is treated with sodium alkoxide to form an ether.

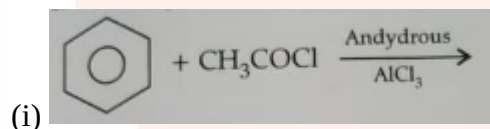


Eg :

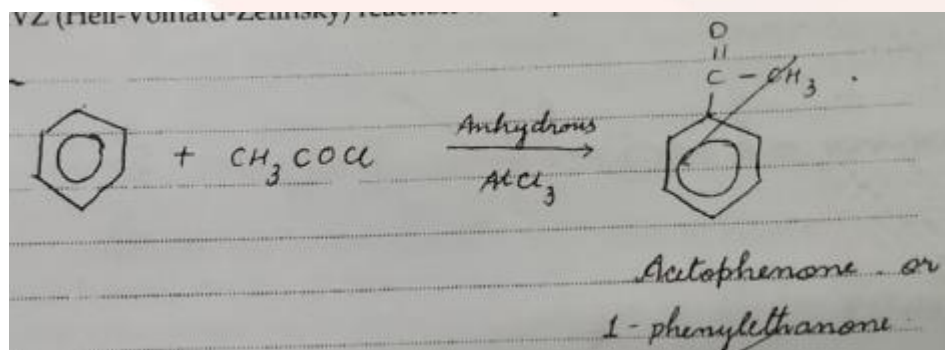


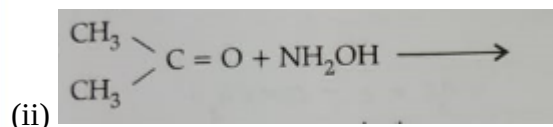
-Ethers containing substituted alkyl groups can be prepared by this reaction.

34. (a) Write the organic compound formed in the following equations.

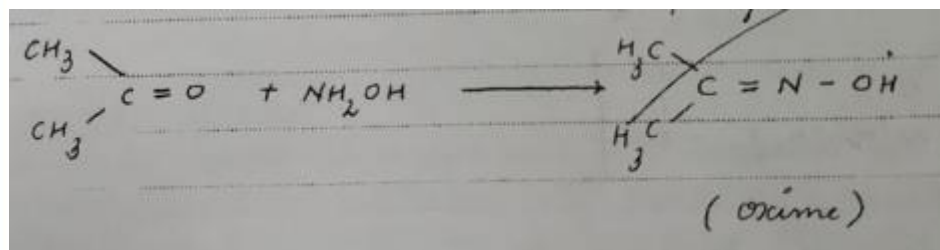


Sol.

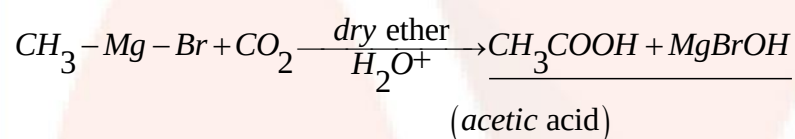




Sol.

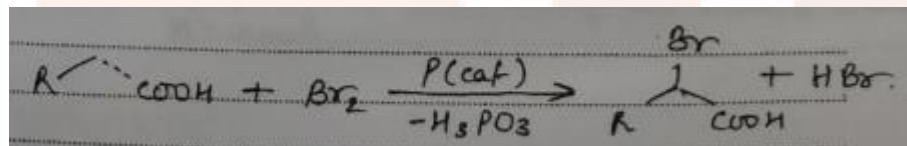


(iii)



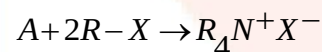
(b) Explain HVZ (Hell-Volhard-Zelinsky) reaction with equation.

Sol. Hell Volhard Zelinsky reaction-



Treatment with bromine and catalytic amount of phosphorus leads to the selective α -bromination of carboxylic acids.

35. (a) identify the reactant 'A' in the following reaction:



(b) Explain Hoffmann's bromamide degradation reaction for the preparation of methanamine.

(c) Which is more basic among aqueous solutions of aniline and ammonia? Give one reason.

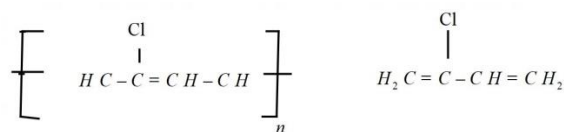
36. (a) Write Haworth structure for maltose.

(b) What is meant by denaturation of proteins? Which level of structure remains intact during denaturation of globular protein?

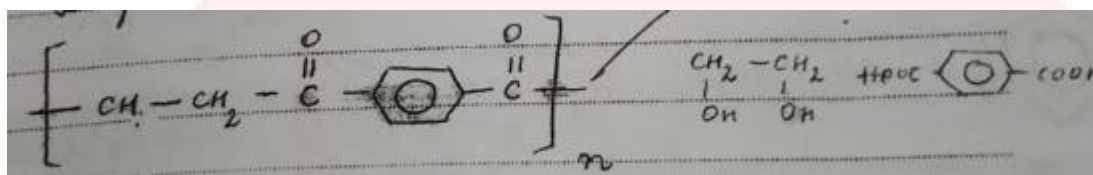
(c) Name the base present only in DNA but not in RNA.

37. (a) Write the partial structure of

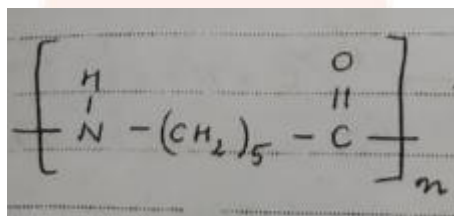
(i) Neoprene



(ii) Terylene(Dacron)

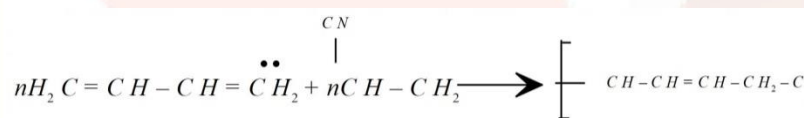


(iii) Nylon-6



(b) Explain the preparation of Buna-N with equation.

Sol. preparation of Buna-N: Buna-N is prepared by the condensation polymerization of 1,3-butadiene and in the presence of acrylonitrile in the presence of a peroxide catalyst.



Buna-N .